

68. [HRPulse Workspace](#)

Internal Human Resource Management Portal

React Final Project

Submitted By:

Name: Samarth Navale

Roll Number: 150096725148

Cohort: Sam Altman

Branch: B.Tech Computer Science Engineering

Semester: 2

Institute: ITM

Academic Year: 2025-2029

Abstract

HRPulse Workspace is a React-based Human Resource Management portal developed to simplify employee administration tasks within an organization. The system provides employee directory management, organizational hierarchy visualization, leave request processing, salary rollback tracking, permission management, reporting chain tracking, tenure analysis, and automated HR task assignment.

The application uses CSV files as the primary data source and operates entirely on the client side without requiring a backend server or database. This approach keeps the system lightweight, portable, and easy to demonstrate while still covering all required HR workflows.

Problem Statement (68. HRpulse Workspace)

Develop an internal company website for HR staff to:

- Manage employee records
 - Track organizational hierarchy
 - Process leave requests
 - Maintain salary change history
 - Assign HR operational tasks
 - Manage role permissions
 - Analyze employee tenure
 - Track reporting relationships
-

Objectives

The objectives of HRPulse Workspace are:

1. Create a centralized HR management dashboard.
2. Provide quick employee lookup and search capabilities.
3. Visualize company hierarchy through an interactive organizational chart.
4. Manage employee salary changes with rollback functionality.
5. Process leave requests using FIFO queue management.
6. Maintain role-based permission visibility.
7. Track reporting chains within the organization.
8. Automatically assign HR tasks to available staff.

Technology Stack

Frontend

- React.js
- JavaScript (ES6+)
- HTML5
- CSS3

Build Tool

- Vite

Data Processing

- PapaParse

UI Icons

- Lucide React

Data Source

- CSV Files

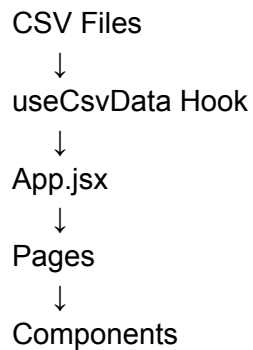
CSV Files Used

1. employees.csv
 2. leaveRequests.csv
 3. salaryHistory.csv
 4. roles.csv
 5. tasks.csv
-

System Architecture

Architecture Overview

The application follows a centralized state architecture.

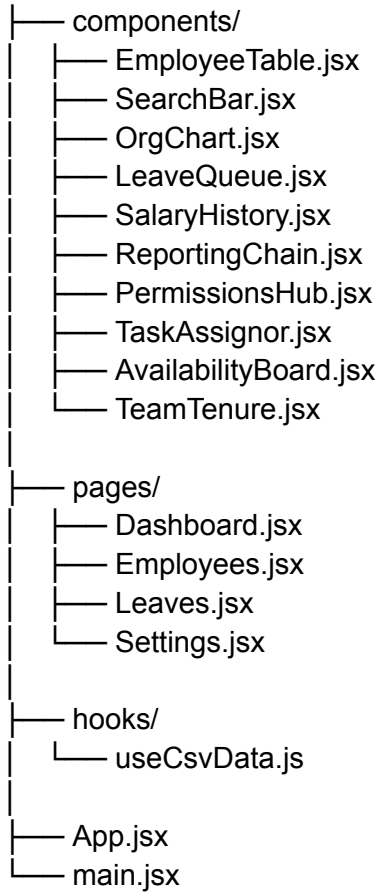


Data Flow

1. CSV files are loaded using PapaParse.
 2. Data is parsed into JavaScript objects.
 3. App.jsx stores all application state.
 4. Components receive data through props.
 5. User actions update state.
 6. Updated state automatically refreshes UI.
-

Project Structure

src/



Features Implementation

A. Company Structure Viewer

Purpose

Display organizational hierarchy in a tree structure.

Implementation

- Recursive TreeNode component
 - Expand/Collapse support
 - Department visibility
 - Manager-Employee relationships
 - Circular reference protection
-

B. Salary Change Undo

Purpose

Track salary modifications and allow rollback.

Implementation

- Salary change logging
 - Previous salary preservation
 - Latest change undo functionality
 - Salary audit history
-

C. Time-Off Request Queue

Purpose

Process leave requests in exact submission order.

Implementation

- FIFO Queue
 - Pending Request List
 - Approve / Reject Actions
 - Processed Request History
 - Processing Timestamp Recording
-

D. Fast Employee Search

Purpose

Quickly locate employee information.

Implementation

- Search by Name
 - Search by Department
 - Search by Role
 - Debounced Search Input
 - Scrollable Employee Directory
-

E. Tenure Sorter

Purpose

Rank departments based on average employee tenure.

Implementation

- Department Grouping
 - Average Tenure Calculation
 - Descending Ranking
 - Team Comparison Table
-

F. Permissions Hub

Purpose

Display and manage role permissions.

Implementation

- Role-Permission Mapping
 - Permission Addition
 - Permission Removal
 - CSV-driven Permission Storage
-

G. Reporting Chain Tracker

Purpose

Identify the reporting path from employee to top management.

Implementation

- Manager Traversal Logic
- Circular Reference Protection
- Management Step Count
- Reporting Path Visualization

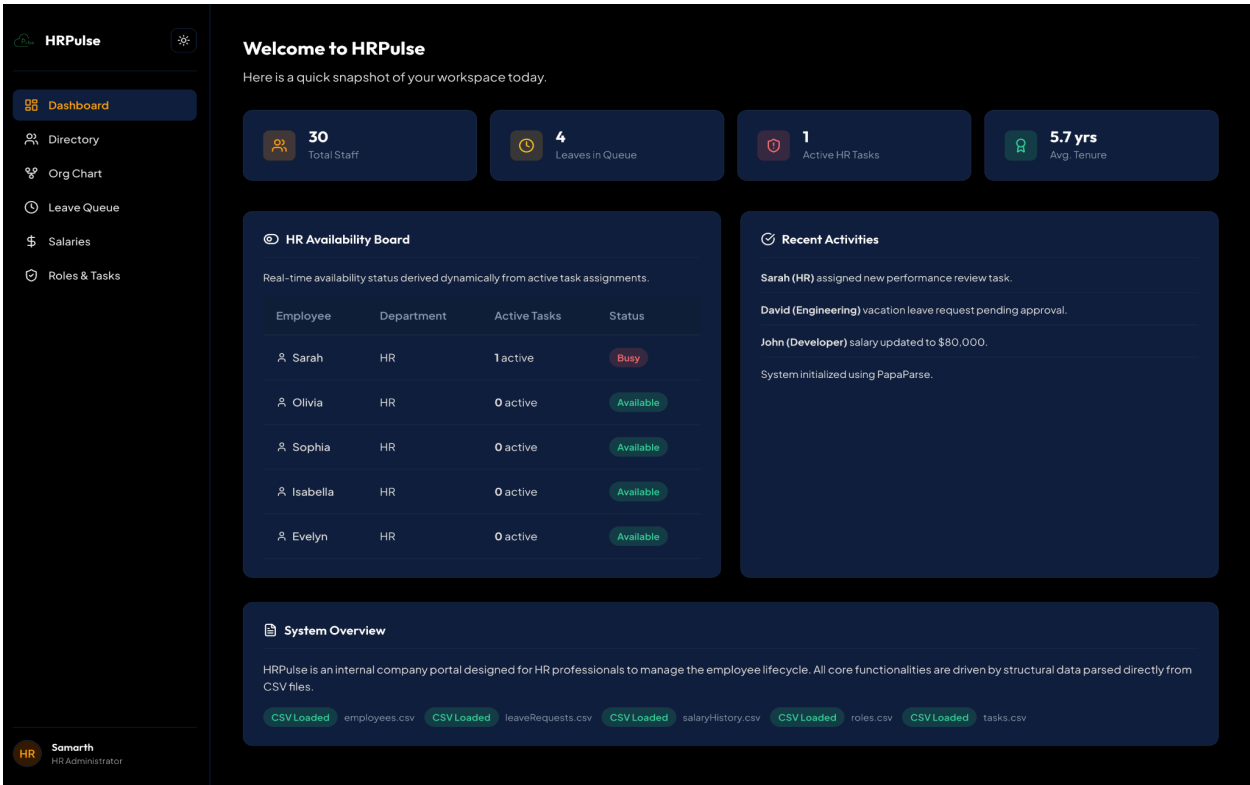
H. HR Task Assignor

Purpose

Automatically distribute HR responsibilities.

Implementation

- HR Availability Detection
- Round Robin Assignment
- Active Task Tracking
- Completed Task Tracking
- Dynamic Availability Calculation



 HRPulse

Dashboard

Directory

Org Chart

Leave Queue

Salaries

Roles & Tasks

HR Administrator

Employee Directory

Search, filter, update salaries, and track report chains.

Active Personnel

Search employee name, department, role...

All Departments

ID	Name	Department	Role	Salary	Join Date	Actions
#1	John	Engineering	Developer	\$80,000	2020-01-15	Update Salary
#2	Sarah	HR	HR Executive	\$60,000	2021-02-10	Update Salary
#3	David	Engineering	Developer	\$85,000	2019-08-12	Update Salary
#4	Alice	Management	CEO	\$150,000	2015-05-10	Update Salary
#5	Marcus	Engineering	Engineering Manager	\$110,000	2017-03-22	Update Salary
#6	Olivia	HR	HR Manager	\$95,000	2018-06-15	Update Salary
#7	Diana	Finance	Finance Manager	\$100,000	2016-11-01	Update Salary
#8	Emily	Engineering	QA Engineer	\$70,000	2021-09-01	Update Salary
#9	Michael	Engineering	Devops Specialist	\$90,000	2018-04-12	Update Salary

Reporting Chain Tracker

Team Tenure Sorter

 HRPulse

Dashboard

Directory

Org Chart

Leave Queue

Salaries

Roles & Tasks

HR Administrator

Reporting Chain Tracker

Find the reporting hierarchy and chain of command from any employee up to the CEO.

Select Employee

-- Choose Employee --

Select an employee above to calculate their reporting chain.

Team Tenure Sorter

Departments ranked by the average time staff members have been with the company.

Rank	Department	Average Tenure	Headcount	Trend
#1	Management	11.1 years	1 employee	High Retention
#2	Operations	6.3 years	3 employees	High Retention
#3	Finance	6.2 years	4 employees	High Retention
#4	HR	5.6 years	5 employees	High Retention
#5	Engineering	5.5 years	13 employees	High Retention
#6	Marketing	4.3 years	4 employees	High Retention

Dashboard

Directory

Org Chart

Leave Queue

Salaries

Roles & Tasks

HR

Samarth

HR Administrator

Company Structure Tree

Interactive reporting viewer. Collapse and expand nodes as needed.

Alice

CEO

Executive

Meera

MANAGEMENT CONSULTANT

Engineering

Chloe

HR MANAGER

HR

Dana

FINANCIAL MANAGER

Finance

Emily

SOFTWARE ENGINEER

Engineering

Michael

DEVOPS SPECIALIST

Engineering

Charlotte

TECHNICAL LEAD

Engineering

Sarah

HR SPECIALIST

HR

Sophia

RECRUITER

HR

Evelyn

HR SPECIALIST

HR

Liam

ACCOUNTANT

Finance

Noah

FINANCIAL ANALYST

Finance

Lucas

CONTENT WRITER

Marketing

AS

MANAGER

Engineering

Henry

MANAGEMENT CONSULTANT

Engineering

Alexander

DEVOPS SPECIALIST

Engineering

Isabella

RECOMMENDATION

HR

Sebastian

SOFTWARE ENGINEER

Engineering

Daniel

SOFTWARE ENGINEER

Engineering

Mia

FINANCIAL ACCOUNTANT

Finance

Dashboard

Directory

Org Chart

Leave Queue

Salaries

Roles & Tasks

HR

Samarth

HR Administrator

Time-Off Requests

Manage the vacation and sick leave queue in submission order.

Active Queue (Pending)

Strict First-In-First-Out (FIFO) queue. Only the front-of-queue request can be processed.

David

Employee ID: #3

#1

Vacation

Submitted: 2025-01-05

Approve

Reject

William

Employee ID: #16

#2

Personal

Submitted: 2025-01-07

Waiting for front item...

Sarah

Employee ID: #2

#3

Maternity

Submitted: 2025-01-08

Waiting for front item...

Sebastian

Employee ID: #24

#4

Sick

Submitted: 2025-01-10

Submit Time-Off Request

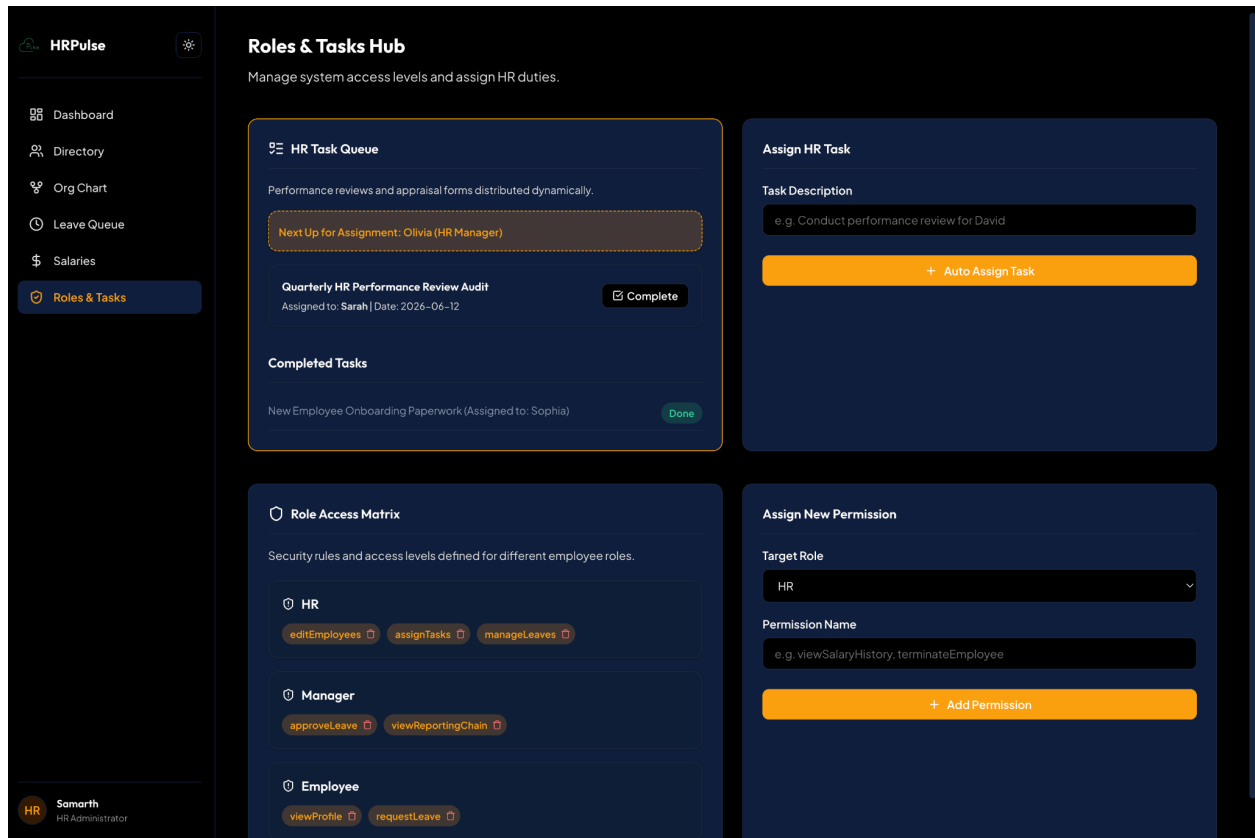
Select Employee

-- Choose Employee --

Leave Type

Vacation

Add to Queue (FIFO)



Future Enhancements

1. Database Integration
 2. User Authentication
 3. Role-Based Access Control
 4. CSV Export Functionality
 5. Employee Profile Editing
 6. Real-Time Notifications
 7. Multi-Manager Reporting Structures
 8. Persistent Data Storage
-

Conclusion

HRPulse Workspace successfully fulfills all specified assignment requirements. The application demonstrates practical usage of React components, state management, CSV-based data handling, recursive rendering, queue management, and derived-state computation.

The project provides a functional HR management portal while maintaining a simple and maintainable architecture suitable for academic evaluation and future expansion.

References

1. React Documentation
2. Vite Documentation
3. PapaParse Documentation
4. Lucide React Documentation

GitHub Repository

GitHub Link: <https://github.com/Samarth-ITM/HRpulse-Workspace>

Live Deployment

Live URL: <https://hrpulse-seven.vercel.app/>
